

A Hybrid Linguistic Time Series Forecasting Model combined with Particle Swarm Optimization

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Abstract— Linguistic time series forecasting model (LTS-FM) which is proposed by Hieu et al. by utilizing hedge algebras theory is an efficient forecasting model. Instead of partitioning the universe of discourse (UD) of the linguistic variable into sub-intervals and assigning fuzzy sets to them, it establishes a formalism to convert historical numeric time series data into linguistic one based on numerical semantics of words which are transformed from the semantically quantifying mapping (SQM) values of the corresponding words. Therefore, a LTS-FM is established in such a way that it handles directly words of linguistic variable and their qualitative semantics. However, the fuzziness parameter values of the LTS-FM which determine the SQM values of words are currently specified by human experts, so the forecasted results may not be optimal. This paper proposes a hybrid LTS-FM in which particle swarm optimization is utilized to optimize the fuzziness parameter values. A new formula of computing crisp forecasted values is also proposed. The experimental studies carried out over two practical forecasting problems of the enrollments of University of Alabama and killed in car road accident in Belgium show that the proposed forecasting model obtains better forecasted results.

Keywords—linguistic time series, LTS, hedge algebras, linguistic logical relationship.

I. INTRODUCTION

Fuzzy time series forecasting model (FTS-FM) proposed by Song and Chissom [1-3] and then enhanced by Chen [4], in which fuzzy sets are considered the computational semantics of words. In that forecasting model, the universe of discourse (UD) of linguistic variable is divided into sub-intervals, fuzzy sets are designed on them, and then linguistic words are assigned to the designed fuzzy sets. The forecasting procedures are established based on fuzzy logical relationships. The studies of them have opened a new intensive research field which has been attracting the research community publishing their researches to enhance the models and solve practical problems. The methodological studies in the literature include simplifying computational method [4], sub-interval length optimization by applying particle swarm optimization (PSO) [5, 16], genetic algorithm (GA) [7], etc. or clustering techniques [26], high-order models [7, 9], multi-factor models [20], application of efficient defuzzification techniques [5, 16], linguistic time series [8, 9, 18, 23], etc. The application-oriented studies focus on the

forecasting problems in practice such as the enrollment forecasting [2-7], stock index forecasting [16, 20], temperature forecasting [20], tourism demand forecasting [21], car road accident forecasting [5, 19], spot gold forecasting [5, 19], etc.

Despite the considerable achievements, there are some drawbacks of FTS-FMs which can briefly be described as follows:

- *The word-semantic problem*: Although the linguistic words present in fuzzy time series with their inherent semantics well-familiar with the human users, they are only considered as linguistic labels intuitively assigned to the designed fuzzy sets. There is no formal connection between the inherent semantics of linguistic labels with their associated fuzzy sets.
- *The determination of the designed fuzzy sets*: It is obvious that the inherent semantic of linguistic labels is objective because all users understand the same its meaning. Meanwhile, the designed fuzzy sets are entirely subjective because they depend very much on the designers.

With the drawbacks described above, the need of a formal formalism that can directly handle linguistic words in solving a given time series forecasting problem is arising.

Hedge algebras (HA) introduced by Nguyen et al. [10, 11] model linguistic variables inherent semantic structures in an axiomatic way. HA establishes formal formalism to handle directly linguistic words and has been applied efficiently in the fields of classification [12, 17], regression [13], fuzzy control [15], time series forecasting [8, 9, 18, 23], image processing [14], and so on. Applied to deal with time series forecasting problems, HA establishes a formalism to convert historical numeric time series into linguistic time series (LTS) based on numerical semantics of words which are computed from the semantically quantifying mapping (SQM) values of the corresponding words. Then, linguistic logical relationships (LLRs) are established from LTS and based on them linguistic logical relationship groups (LLRGs) are established. They are the basis for establishing forecasting rules. However, we can see that although the LTS-FMs proposed in [8, 9, 18, 23] handle directly words of linguistic variable, they are still restricted more or less in the methodological frame of FTS-FMs. Besides, the

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fuzziness parameter values of those LTS-FMs which determine the SQM values of words are currently specified by human experts. Therefore, the generated LTS may not be optimal leading to the forecasting accuracies may not be good enough.

This paper proposes a method of optimizing the fuzziness parameter values of LTS-FMs by utilizing particle swarm optimization (PSO) [22] so as to improve their forecasting accuracies. A new formula of computing crisp forecasted values based on local and global information is also proposed. The experimental studies carried out over two real forecasting problems of the enrollments of University of Alabama observed from 1971 to 1992 (EoU of Alabama) and killed in car road accident in Belgium observed from 1974 to 2004 (CRA in Belgium) show that the proposed forecasting method obtains better forecasted results.

The rest of the paper is organized as follows. In Section II, we recall some related basic knowledge. Section III presents our proposed LTS-FM. Some experimental studies are presented in Section IV. Some conclusions are drawn out in Section V.

II. PRELIMINARIES

A. Fuzzy Time Series and Some Related Concepts

Because the LTF-FM was formed inspired by the concept of fuzzy time series (FTS), for comparison purpose between LTS-FM and FTS-FM and to make this study self-contained, some basic concepts of FTS will be briefly summarized.

In [1], Song & Chissom introduced the concept of FTS as follows.

Definition 1. Let $X(t) (t = 0, 1, 2, \dots)$, which is the subset of real numbers, be the universe of discourse on which fuzzy sets $f_i(t) (i = 1, 2, \dots)$ are defined. If $F(t)$ is the collection of $f_i(t) (i = 1, 2, \dots)$ then $F(t)$ is called a fuzzy time series on $X(t) (t = 0, 1, 2, \dots)$ [1, 2].

Definition 2. Suppose $F(t)$ is caused by $F(t-1)$, denoted by $F(t-1) \rightarrow F(t)$, then this relationship can be represented as the equation $F(t) = F(t-1) \circ R(t, t-1)$, where $R(t, t-1)$ is a fuzzy relationship between $F(t-1)$ and $F(t)$ and ' $\circ$ ' is a composition operator [1-3].

Let $F(t) = A_j$, $F(t-1) = A_i$, we have the relationship $A_i \rightarrow A_j$ called fuzzy logical relationship (FLR), where A_j is the right-hand side and A_i is the left-hand side of the FLR [4].

Definition 3. Suppose there are FLRs $A_i \rightarrow A_{j1}$, $A_i \rightarrow A_{j2}$, ..., $A_i \rightarrow A_{jn}$ with the same left-hand side. Then, they can be grouped into a fuzzy logical relationship group (FLRG), denoted by $A_i \rightarrow A_{j1}, A_{j2}, \dots, A_{jn}$ [4].

The MSE (mean square error) value is used to evaluate the forecasting models and is calculated as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (F_i - A_i)^2, \quad (1)$$

where F_i is the forecasted data, A_i is the actual data, and n is the number of forecasted data.

In FTS-FM, the authors considered the observations as linguistic variables where their values are represented by fuzzy

sets labelled by linguistic words. In [8], Hieu et al. proposed a formalism for directly handling words of linguistic variables.

B. Hedge Algebras - mathematical models of linguistic variables

As aforementioned, the use of fuzzy sets in representing semantics of linguistic values has some drawbacks. Hedge algebras, which was introduced by N.C. Ho and Wechler [10, 11], can provide a solid mathematical foundation to formalize the word-domain of linguistic variables efficiently.

Let $\mathcal{X}$ be a linguistic variable. A hedge algebras (HA) of $\mathcal{X}$, denoted by $\mathcal{AX}$, be the 5-tuple $(X, G, C, H, \leq)$. In this tuple, X is the word-domain of $\mathcal{X}$; $\leq$ is the order relationship on X ; $G = \{c^-, c^+\}$ is the set of atomic antonym called generators. For example, if $\mathcal{X} = AGE$ is the age of a person, two its elements 'old' and 'young' are positive and the negative atomic words, respectively. H is a set of adverbs called linguistic hedges which are considered as unary operators. For instance, $H^- = \{Rather\}$ and $H^+ = \{Very\}$ can be two opposite hedges which have capacity to increase and decrease meaning of any word in X , i.e., *Rather old* $\leq$ *old* and *old* $\leq$ *Very old*. $C = \{0, W, 1\}$ is the set of three constants, i.e., 'Absolutely young', 'middle' and 'Absolutely old' for the linguistic variable *AGE*. Hereafter, we summarize some fundamental concepts of HA related to the LTS-FM.

Definition 4. Let $\mathcal{AX} = (X, G, C, H, \leq)$ be a HA. A function $m: X \rightarrow [0, 1]$ is called 'fuzziness measure' on X if:

- $m(0) = m(W) = m(1) = 0$;
- $m(c^-) + m(c^+) = 1$;
- $\sum_{h \in H} m(hu) = m(u)$, for $\forall u \in X$;
- For $\forall x, y \in X, \forall h \in H, \frac{m(hx)}{m(x)} = \frac{m(hy)}{m(y)}$. This proportion is called *fuzziness measure of the hedge h* and denoted by $\mu(h)$ for simplicity.

The properties of fuzziness measure m of $\mathcal{X}$ are as follows:

- f1) $m(hx) = \mu(h)m(x)$, for $\forall x \in X$;
- f2) $\sum_{-q \leq i \leq p, i \neq 0} m(h_i x) = m(x)$, $\forall x \in X$;
- f3) $\alpha = \sum_{-q \leq i \leq -1} \mu(h_i)$, $\beta = \sum_{1 \leq i \leq p} \mu(h_i)$, where $\alpha, \beta > 0$ and $\alpha + \beta = 1$, called the total of fuzziness measure of negative hedges and positive hedges, respectively.

For $\forall x \in X$, there is a mapping called *sign* function, which determine relations before and after the hedge impact to a specific word. This mapping is defined as follows.

Definition 5. A sign function is a mapping **sign**: $X \rightarrow \{-1, 1\}$ defined as follows. For $h, h' \in H$ and $c \in \{c^-, c^+\}$:

- 1) $sign(c^-) = -1$, $sign(c^+) = +1$;
- 2) $sign(hc) = -sign(c)$ ($= +sign(c)$) if h is negative (positive) w.r.t c ;
- 3) $sign(h'hx) = -sign(hx)$ ($= +sign(hx)$) if $h'hx \neq hx$ and h' is negative (positive) w.r.t h ;

In the LTS-FM, the quantitative semantic value of used words plays an important elements. For a specific HA, when all the fuzziness parameter values of the variable are given, the quantitative semantics of all linguistic values are calculated as the following definition.

Definition 6. The semantically quantifying mapping $v: X \rightarrow [0, 1]$ is defined as follows:

- 1) $v(W) = \theta = m(c^-)$;
- 2) $v(c^-) = \theta - \alpha m(c^-)$ and $v(c^+) = \theta + \alpha m(c^+)$;
- 3) $v(h_j x) = v(x) + \text{sign}(h_j x) \sum_{i=\text{sign}(j)}^j m(h_i x) - \omega(h_j x) m(h_j x)$
where $\omega(h_j x) = \frac{1}{2} [1 + \text{sign}(h_j x) \text{sign}(h_p h_j x) (\beta - \alpha)] \in \{\alpha, \beta\}$.

Once all the fuzziness parameter values are given, the semantically quantifying mapping (SQM) value of words are determined. Then, using the linguistic time series concepts, a forecasting model will be established [8, 9]. In the next section, we will present some LTS concepts and its forecasting model.

C. Linguistic Time Series and its Forecasting Model

In the real world, humans use language as a means of communication. They also use the linguistic values as the labels to collect, store, and process their observed data. Language processing in the human brain is a complex process. But in their lifetime, sometimes the linguistic information was natural collected as a LTS which can be formally defined as follows.

Definition 7 [8]. Let $\mathbb{X}$ be a set of linguistic values of a linguistic variable $\mathcal{X}$ defined on the universe of discourse U_x to describe its numeric quantities. Then, any series $L(t)$, $t = 0, 1, 2, \dots$, where $L(t)$ is a collection of words of $\mathbb{X}$, is called a linguistic time series.

For instance, observing a person's mood at equidistant time intervals, we can obtain linguistic time series such as "good", "Very good", "Rather good", "normal", "bad", "Rather bad", "bad", "Very bad" and so on. This linguistic time series was natural built when humans use English to collect the data.

Because humans collect linguistic data in the real life in the chronological order, the time order relationships between linguistic values also exist naturally. We call this kind of relationship *linguistic logical relationships* (LLR) and defined as follows.

Definition 8. Suppose X_i and X_j are the linguistic values which represent the data at the time t and $t + 1$, respectively. There exists a relationship between X_i and X_j , and it is called linguistic logical relationship (LLR), denoted by $X_i \rightarrow X_j$.

When collecting historical data, there are many relationships that arise from one linguistic value can create many other linguistic value. In other word, these relationships have the same left-hand side. Then, we can then group the LLRs into LLRG as defined below.

Definition 9. Suppose there are LLRs such as $X_i \rightarrow X_{j1}$, $X_i \rightarrow X_{j2}$, ..., $X_i \rightarrow X_{jn}$ which have the same left-hand side. They are grouped into a linguistic relationship group (LLRG), denoted by

$$X_i \rightarrow X_{j1}, X_{j2}, \dots, X_{jn}.$$

The forecasting model for linguistic time series is demonstrated as in Fig. 1.

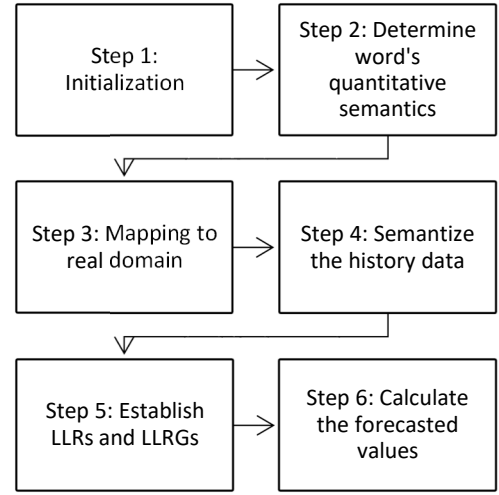


Fig. 1. The linguistic time series forecasting model

The forecasting procedure of LTS-FM with six steps is described as follows.

Step 1. Specify the UD. Establish HA with given fuzzy parameter values and appropriate linguistic values (words for simplicity).

Step 2. Generate the SQM values of the used words using Def. 6.

Step 3. Map the SQM values to the UD, then we get a collection of real semantics of words.

Step 4. Semantize the historical time series data. Use the minimum distance from a specific datum to the real semantics to determine the word label of this datum.

Step 5. Generate the LLRs and LLRGs between used words.

Step 6. Forecast based on LLRGs and calculate the crisp forecasted values by the principles as follows.

- (1) If linguistic value at a forecasting time, assume k , is X_i and its right-hand side is $X_i \rightarrow X_{j1}, X_{j2}, \dots, X_{jp}$, $p \geq 1$, the crisp forecasted value at time $k + 1$ is $(s_{j1} + s_{j2} + \dots + s_{jp})/p$ where $s_{j1}, s_{j2}, \dots, s_{jp}$ are the real semantic (s) of $X_{j1}, X_{j2}, \dots, X_{jp}$, respectively.
- (2) If linguistic value at a forecasting time, assume k , is X_i and its right-hand side is empty, the crisp forecasted value at time $k + 1$ is s_i , where s_i is the real semantic of X_i .

To have more information for the forecasting process, a new formula of crisp forecasted values is proposed to applied. It is the integration of the local and the global information to enhance forecasted accuracy of our proposed model. Therefore, the forecasted value of year $k + 1$ in principle (1) in Step 6 is computed as: $0.5 \times (s_r + (s_{j1} + s_{j2} + \dots + s_{jp})/p)$, where s_r is the real semantic of word in the right-hand side of the LLRG.

D. Particle Swarm Optimization

Particle swarm optimization (PSO) was introduced by Kennedy and Eberhart in 1995 [22]. It really proved to be an

efficient optimization algorithm and has been applied to solve many real world problems. The parameters of basic PSO is described as follows:

A swarm $S = \{x_1, x_2, \dots, x_N\}$ has N particles and each particle x_i has its position X_i^t at generation t updated as follows:

$$X_i^{t+1} = X_i^t + V_i^{t+1} \quad (2)$$

where V_i^{t+1} is the velocity at generation $t + 1$ of particle x_i and it is updated as follows:

$$V_i^{t+1} = \omega V_i^t + c_1 r_1 (P_i^t - X_i^t) + c_2 r_2 (P_g^t - X_i^t), \quad (3)$$

where P_i^t and P_g^t are the local solution and the best global solution found so far; two random numbers r_1 and r_2 are distributed uniformly in $[0, 1]$; c_1 , c_2 and ω are self-cognitive factor, social cognitive factor and inertia weight, respectively. The basic PSO procedure is briefly described as follows:

Step 1: Initialize swarm S which has its velocity vector V and position vector X are randomly generated, the variable t indicating the current cycle step, and the constant G_{max} indicating the number of cycles.

Step 2: Compute new value of $f(X_i^t)$ which is the objective of particle x_i .

Step 3: Compare $f(X_i^t)$ with the old one of each particle x_i . If $f(X_i^t)$ is better, then update P_i^t with the current position of x_i .

Step 4: Update the global best position P_g^t if there exists $f(X_i^t)$ better the one of all particles in the swarm and the one of P_g^t .

Step 5: Update V_i^t by Eq. (2) and X_i^t by Eq. (3).

Step 6: Increase variable t and repeat from Step 2 until the value of t greater than the number of cycles G_{max} .

III. A HYBRID FORECASTING MODEL WITH THE COMBINATION OF LTS AND PSO

In this section, a hybrid LTS-FM combined with PSO is presented. In the FTS-FMs, the interval lengths of the UD are optimized to improve the forecasted accuracies. In LTS-FMs, the words of linguistic variable X induced by the associated hedge algebras are used to describe the time series data, the fuzziness parameter values including a fuzziness measure of a generator word and linguistic hedges are used to generate real numerical semantics, based on which the linguistic words are assigned to time series data. Therefore, the values of fuzziness parameters are very important and should be optimized to improve forecasted values. In our experiments, as in [10, 11], only two linguistic hedges (*Little* and *Very*) are used and the number of linguistic words is fixed for each model, so the number of the optimized fuzziness parameter values is only 2. More specifically, they are the fuzziness measure of one of two generator words (e.g. $m(c^-)$) and the one of the hedge *Little* or *Very* (e.g. $\mu(Little)$). We have $m(c^-) + m(c^+) = 1$ and $\mu(Little) + \mu(Very) = 1$, so we can infer that $m(c^+) = 1 - m(c^-)$

) and $\mu(Very) = 1 - \mu(Little)$. Each particle position $X_i^t = \{m(c^-), \mu(Little)\}$ ($i = 1, \dots, N$) in swarm represents those two fuzziness parameter values based on which a forecasting procedure is implemented. The details of each step of our proposed forecasting model are as follows:

Algorithm 1. LTSPSO.

Input: The historical data of time series D ;

Parameters: N , G_{max} , the syntactic semantics of the associated hedge algebras HA ;

// N is the swarm size, G_{max} is the number of cycles.

Output: The best fuzziness parameter values $P_g^{G_{max}-1}$;

Begin

Randomly initialize a swarm $S_0 = \{X_i^0; V_i^0 \mid i = 1, \dots, N\}$, where $X_i^0 = \{m(c^-), \mu(Little)\}$;

For $i = 1$ to N do begin

Implement Step 2 to 6 of LTS-FM based on X_i^0 ;

Compute MSE value by the formula (1);

Update the personal best P_i^0 of the particle x_i based on MSE value;

End;

Update the global best position P_g^0 based on MSE values;
 $t = 1$;

Repeat

For $i = 1$ to N do begin

Update velocity V_i^t of particle x_i using (3);

Update new position X_i^t of particle x_i using (2);

Implement Step 2 to 6 of LTS-FM based on X_i^t ;

Compute MSE value of particle x_i by formula (1);

If X_i^t is better than X_i^{t-1} **then**

Update personal best P_i^t of the particle x_i ; based on MSE value;

End;

End;

Update the global best position P_g^t based on MSE values;

$t = t + 1$;

Until $t = G_{max}$;

Return the best solution $P_g^{G_{max}-1}$;

End.

After the optimization process, we receive the best fuzziness parameter values ($m(c^-)$ and $\mu(Little)$) which are used to convert numeric time series data into linguistic one.

IV. EXPERIMENTAL STUDIES AND DISCUSSION

The aim of this section is to present the experimental studies of our proposed forecasting model. The experimental results the proposed forecasting models are compared with the existing ones to show the outperformance of them.

A. Experiment Setup

All our experiments are implemented by C# programming language. The experiments are executed on Windows 10, laptop

with Core i5-8250U 1.6 GHz processor, RAM 8GB. The parameter values of PSO are as follows: the number of cycles: 1000, the number of particles: 300, Inertia coefficient: 0.4, $c1 = c2 = 2.0$.

Each forecasting model applied to solve a given forecasting problem is executed three times, so we get three corresponding MSE values and the smallest one will be selected for experimental result.

B. Forecast Enrollments of University of Alabama

In this forecasting problem, the syntactical semantics of linguistic variable $\mathcal{X}$ are given as follows:

- Two generator $c^- = \text{small} (s)$ and $c^+ = \text{large} (l)$ and two hedges *Little* (L) $\in H^-$, *Very* (V) $\in H^+$.
- The used word-set is selected from $X_{(3)}$, so have: $X_L = \{\theta, VV_small, V_small, LV_small, small, LL_small, L_small, VL_small, medium, VL_large, L_large, LL_large, large, LV_large, V_large, VV_large, I\}$, where θ and I stand for *Absolutely small* and *Absolutely large*, respectively.
- The historical data of the EoU of Alabama which is observed from 1971 to 1992 is ranged from 13055 to 19337, so the UD of $\mathcal{X}$ is set to [13000, 20000].

Apply the proposed algorithm LTSPSO to optimize the fuzziness parameter values for LTS-FMs, we have $m(c^-) = 0.4789$ and $\mu(L) = 0.608$, the corresponding MSE value is 22403, much better than the one of Hieu et al. in [8, 9].

TABLE I. THE FORECASTED VALUES OF THE EoU OF ALABAMA OF OUR PROPOSED MODEL IN COMPARISON WITH THE EXISTING ONES

Year	Actual	Uslu	Chen&Zou	Proposed
1971	13055			
1972	13563	13650	13469	13515
1973	13867	13650	13952	14001
1974	14696	14836	14596	14800
1975	15460	15332	15439	15509
1976	15311	15447	15241	15509
1977	15603	15447	15925	15509
1978	15861	15447	15880	15752
1979	16807	16746	16801	16693
1980	16919	17075	17009	16949
1981	16388	16380	16260	16779
1982	15433	15504	15435	15553
1983	15497	15431	15212	15509
1984	15145	15077	15282	15132
1985	15163	15297	15344	15132
1986	15984	15848	15714	15752
1987	16859	16835	16833	16693
1988	18150	18145	18016	17888

Year	Actual	Uslu	Chen&Zou	Proposed
1989	18970	18880	18937	18911
1990	19328	19418	19345	19439
1991	19337	19260	19147	19307
1992	18876	19031	19152	19043
MSE		42250	23710	22403

To show the efficiency of our proposed model, we do compare its forecasted results with the ones of the existing LTS-FMs. Table I shows the experimental results of the EoU of Alabama of the proposed forecasting model in comparison with the forecasting models of Uslu et al. [19] and Chen & Zou [5]. We can see that the MSE value of our proposed forecasting model is 22403, less than the ones of Uslu and Chen & Zou which are 42250 and 23710, respectively. Therefore, we can state that our proposed forecasting model is very efficient with the enrollment forecasting problem.

C. Forecast Car Road Accident in Belgium

The syntactical semantics of linguistic variable $\mathcal{X}$ of this forecasting problem are as follows:

- Two generator $c^- = \text{low} (l)$ and $c^+ = \text{high} (h)$ and two hedges *Little* (L) $\in H^-$, *Very* (V) $\in H^+$.
- The used word-set is of the form $X_{(3)}$, so have: $X_L = \{\theta, VV_low, V_low, LV_low, low, LL_low, L_low, VL_low, medium, VL_high, L_high, LL_high, high, LV_high, V_high, VV_high, I\}$, where θ and I stand for *Absolutely low* and *Absolutely high*, respectively.
- The UD of $\mathcal{X}$ of the forecasting problem of CRA in Belgium is set to [900, 1700].

TABLE II. THE COMPARISON OF MSE VALUE BETWEEN THE PROPOSED MODEL AND THE COUNTERPARTS

Forecasting model	Uslu	Chen&Zou	Proposed
MSE	1731	1024	740

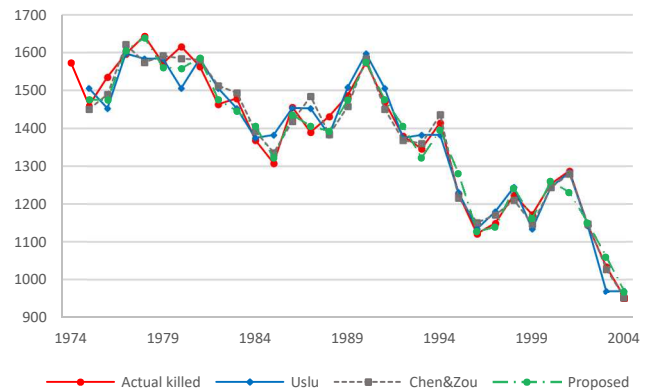


Fig 2. The comparison of car road accident forecasted results

We also apply the proposed algorithm LTSPSO to optimize the fuzziness parameter values for LTS-FMs when applied to the problem of CRA in Belgium. After the optimization process, we get $m(c^-) = 0.4637$ and $\mu(L) = 0.5713$. The MSE values of our

proposed forecasting model and their counterparts, Uslu et al. [19] and Chen & Zou [5], are shown in Table II. The forecasted results in detail are visualized in Fig. 2. It is easy to see that the MSE value of our proposed forecasting model is 740 which is much less than the ones of their counterparts, in turn, of 1731 and 1024. With these experimental results, we conclude that our proposed forecasting model is also very efficient with the forecasting problem of CRA in Belgium.

V. CONCLUSIONS

The LTS-FMs are developed by utilizing the formalism of directly computing with words based on hedge algebras. However, the fuzziness parameter values of the existing LTS-FMs are specified by human experts, so their forecasted accuracies are not quite good. To increase forecasted accuracy, this paper presents a method of optimizing the fuzziness parameter values of LTS-FMs in order to improve their forecasted results. Besides, a new method of computing the crisp forecasted results is also proposed. The experimental results over two forecasting problems, EoU of Alabama and CRA in Belgium, show that our proposed LTS-FM outperforms their counterparts. However, there are some characteristics of LTS-FMs needed to be examined such as the number of used words, the word semantic aspects, etc. They will be examined in our next studies.

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